

IMPLEMENTATION AND VALIDATION GUIDE

VanFox Active Directory Lab

Step-by-Step Implementation, Configuration, and Validation

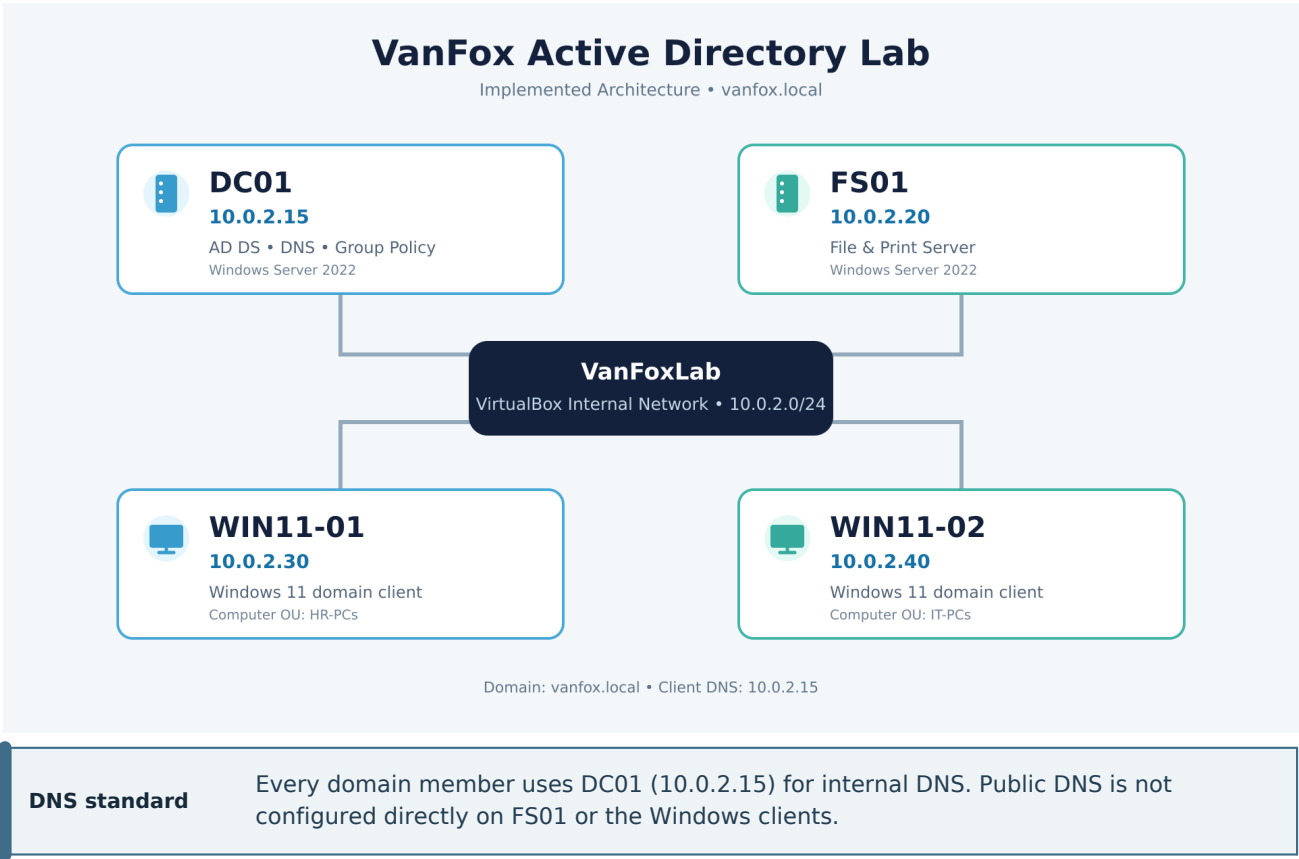
LAB OWNER	DOMAIN	STATUS
Van Neih Thawng	vanfox.local	Complete and validated

A practical four-VM Windows domain portfolio project covering identity, DNS, organizational units, role-based access, file and print services, user-data protection, Group Policy, endpoint administration, and repeatable validation.

Project Overview

The VanFox lab simulates a small-business Windows environment on an isolated Oracle VirtualBox network. The completed environment centralizes identity, name resolution, storage, security policy, software, printing, and Windows client management.

Component	Final configuration
Platform	Oracle VirtualBox - internal network VanFoxLab
Domain	vanfox.local
Domain controller	DC01 - Windows Server 2022 - 10.0.2.15/24
File and print server	FS01 - Windows Server 2022 - 10.0.2.20/24
HR client	WIN11-01 - Windows 11 Enterprise Evaluation - 10.0.2.30/24
IT client	WIN11-02 - Windows 11 Enterprise Evaluation - 10.0.2.40/24
Identity scope	20 users - 5 departments - 10 custom GPOs



Implementation Sequence

Phase	Repository section	Primary result
01	Infrastructure	Four VMs on the isolated VanFoxLab network
02	Domain Controller Promotion	AD DS, DNS, and vanfox.local
03	OU Design	Department, computer, and administrative hierarchy
04	Security Groups	Group-based access and policy targeting
05	User Provisioning	20 realistic department accounts
06	File Server	Secured SMB storage and print services
07	Personal Home Folders	Private H: drive for each user
08	Folder Redirection	Documents stored centrally on FS01
09	Logon Script	PowerShell session marker deployed by GPO
10	Group Policy	Security, resources, and endpoint configuration
11	Client Join and Validation	Two managed clients and end-to-end testing

Final Design Decisions

Decision	Implemented standard	Reason
Personal storage	H: -> \\FS01\Home\$\%USERNAME%	Private per-user storage
Department storage	S: -> matching department share	Avoids conflict with the Home drive
Public storage	P: -> \\FS01\Public	Consistent read-only shared location
Policy structure	Separate VC- GPOs by purpose	Simpler targeting and troubleshooting
Access assignment	AD security groups	Least privilege and reusable administration

Guide alignment

The GitHub repository uses 11 phases. Validation, troubleshooting, and portfolio evidence are integrated into those phases instead of being separate phases 12 and 13.

01

Infrastructure

A repeatable four-system lab foundation

Objectives

- Create separate server and client virtual machines.
- Place every VM on the same isolated internal network.
- Use computer names that match the intended Windows hostnames.
- Capture a clean checkpoint before major role or domain changes.

Final VM Inventory

VM	Operating system	Role	IPv4
DC01	Windows Server 2022	AD DS and DNS	10.0.2.15
FS01	Windows Server 2022	File and Print Server	10.0.2.20
WIN11-01	Windows 11 Enterprise Evaluation	HR domain client	10.0.2.30
WIN11-02	Windows 11 Enterprise Evaluation	IT domain client	10.0.2.40

Procedure

- 1 Create DC01, FS01, WIN11-01, and WIN11-02 in Oracle VirtualBox.
- 2 Attach all network adapters to the internal network named VanFoxLab.
- 3 Allocate sufficient CPU, memory, and disk for Windows updates and role installation.
- 4 Install Windows and rename each operating system to match the VM name.
- 5 Create a clean checkpoint before installing AD DS or joining the domain.

Validation

- All four VMs appear in the VirtualBox inventory.
- All four systems share the 10.0.2.0/24 internal network.
- Server and client roles remain separated.

02

Domain Controller Promotion

Centralized identity and internal DNS

Setting	Final value
Server	DC01
Address	10.0.2.15/24
Roles	Active Directory Domain Services and DNS Server
Forest/domain	vanfox.local
DNS zone	AD-integrated forward lookup zone for vanfox.local

Procedure

1	Install Windows Server 2022, apply updates, rename the server DC01, and restart.
2	Assign static IPv4 address 10.0.2.15/24.
3	Install the AD DS and DNS Server roles.
4	Promote DC01 by selecting Add a new forest and creating vanfox.local.
5	After promotion, configure DC01 to use 10.0.2.15 as preferred DNS.
6	Verify host records for DC01 and FS01 in the vanfox.local forward lookup zone.

Validation Commands

```
dcdiag
nslookup dc01.vanfox.local
nslookup fs01.vanfox.local
```

- Active Directory Users and Computers displays vanfox.local.
- DNS Manager displays the AD-integrated forward lookup zone.
- DC01 resolves to 10.0.2.15 and FS01 resolves to 10.0.2.20.

DNS lesson

A 'Server: Unknown' line in nslookup normally indicates that the DNS server address has no reverse PTR record. Forward resolution can still work, but adding a reverse lookup zone improves diagnostic clarity.

03

OU Design

A policy-ready directory hierarchy

Create the root OU VanFox and separate users, computers, groups, and administrative objects. Department OUs receive user policies, while child computer OUs support device-targeted policies.

Parent	Child OU	Purpose
VanFox	_Admin	Administrative objects
_Admin	Groups	Central security-group location
VanFox	IT, HR, Finance, Operations, Warehouse	Department user accounts
VanFox	Computers	Parent for managed endpoints
Computers	IT-PCs, HR-PCs, Finance-PCs, Operations-PCs, Warehouse-PCs	Department-specific computer targeting

Procedure

1	Create the VanFox root OU in Active Directory Users and Computers.
2	Create the five department user OUs.
3	Create _Admin and its Groups child OU.
4	Create Computers and the five department computer OUs.
5	Move WIN11-01 into HR-PCs and WIN11-02 into IT-PCs after domain join.

Validation

- User accounts are stored in department OUs rather than default containers.
- Security groups are centralized in VanFox_Admin\Groups.
- WIN11-01 is in HR-PCs and WIN11-02 is in IT-PCs.

Design principle	OU placement controls administration and GPO scope; security groups control access. Do not use OUs as a replacement for resource-permission groups.
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04

Security Groups

Reusable role and resource access control

Group	Purpose
HR_Share_RW	Modify access to the HR share
Finance_Share_RW	Modify access to the Finance share
Operations_Share_RW	Modify access to the Operations share
Warehouse_Share_RW	Modify access to the Warehouse share
Public_Share_R	Read access to the Public share
IT_Admins	Authorized local and domain administration
IT_Helpdesk	Delegated support role
Folder_Redirection_Users	Folder Redirection security filtering
USB_Allowed	Approved removable-storage exception
Printer_Users	Shared-printer targeting

Implementation Standard

- Create each object as a Global Security group in VanFox_Admin\Groups.
- Grant share and policy access to groups, not directly to users.
- Separate administrative groups from department resource groups.
- Require users to sign out and sign in after important membership changes.

Representative Role Assignment

User	Selected groups	Reason
Sarah Lee (slee)	IT_Admins, Printer_Users, USB_Allowed, Public_Share_R, Folder_Redirection_Users	Systems administrator and authorized policy exception user
Anna Brown (abrown)	HR_Share_RW, Public_Share_R, Folder_Redirection_Users	HR manager and non-exempt USB test user

05

User Provisioning

Twenty realistic department identities

Department	Name	Username	Role
IT	David Kim	dkim	IT Manager
IT	Sarah Lee	slee	Systems Administrator
IT	Michael Johnson	mjohnson	IT Support Specialist
IT	Emily Chen	echen	Helpdesk Technician
IT	Alex Rivera	arivera	Junior IT
HR	Anna Brown	abrown	HR Manager
HR	Lisa Wong	lwong	Recruiter
HR	Kevin Scott	kscott	HR Assistant
Finance	Maria Lopez	mlopez	Finance Manager
Finance	Robert Davis	rdavis	Accountant
Finance	James Wilson	jwilson	AP Specialist
Finance	Olivia Turner	oturner	AR Specialist
Operations	Daniel Harris	dharris	Operations Lead
Operations	Sophia Clark	sclark	Coordinator
Operations	Mark Evans	mevans	Planner
Operations	Chloe Reed	creed	Analyst
Operations	Henry Brooks	hbrooks	Scheduler
Warehouse	Brian King	bking	Warehouse Supervisor
Warehouse	Jason Young	jyoung	Forklift Operator
Warehouse	Eric Hall	ehall	Picker

Account Standard

- Use the listed username for both sAMAccountName and the user logon name.
- Populate display name, department, job title, and a Department - Job Title description.
- Add all users to Public_Share_R and Folder_Redirection_Users.
- Add users to the matching department share group and additional policy groups according to role.
- Never store passwords, product keys, tokens, or personal information in repository evidence.

06

File and Print Server

Centralized storage, access control, and shared resources

Setting	Final configuration
Server	FS01 - Windows Server 2022 - 10.0.2.20/24
DNS	10.0.2.15 (DC01 only)
Membership	Domain member of vanfox.local
Data volume	Dedicated 40 GB GPT/NTFS disk - D: - label Data
Roles	File Server and Print Server

Share Design

Share	Local path / source	Target group	Access
HR	D:\Shares\HR	HR_Share_RW	Change / Modify
Finance	D:\Shares\Finance	Finance_Share_RW	Change / Modify
Operations	D:\Shares\Operations	Operations_Share_RW	Change / Modify
Warehouse	D:\Shares\Warehouse	Warehouse_Share_RW	Change / Modify
Public	D:\Shares\Public	Public_Share_R	Read / Read & Execute
Home\$	D:\Shares\Home	Per-user ACL	Private
Redirected\$	D:\Shares\RedirectedFolders	Folder_Redirection_Users	Private
Software\$	Chrome MSI package source	Managed computers	Read

Procedure

1	Install Windows Server 2022, rename the server FS01, and assign 10.0.2.20/24.
2	Configure preferred DNS as 10.0.2.15, join vanfox.local, and restart.
3	Attach and format the dedicated 40 GB data disk as D: with NTFS.
4	Install File Server and Print Server roles using an authorized administrator account.
5	Create the D:\Shares structure and SMB shares.
6	Remove broad Everyone permissions; grant access through domain security groups.
7	Enable Access-Based Enumeration and disable offline caching for department shares.

Permission model

Effective access is the most restrictive combination of share and NTFS permissions. Domain Admins, local Administrators, and SYSTEM retain the administrative permissions required for management and recovery.

07

Personal Home Folders

Private and persistent H: storage for each user

Setting	Final value
Server root	D:\Shares\Home
Hidden share	\\FS01\Home\$
Per-user path	\\FS01\Home\$\%USERNAME%
Drive letter	H:
Permissions	User: Modify; Domain Admins and SYSTEM: Full Control

Procedure

1	Create D:\Shares\Home and share it as Home\$.
2	Create one subfolder for every username.
3	Disable inherited permissions on each user folder while preserving required administrative entries.
4	Grant the matching user Modify access only to that user's folder.
5	In ADUC user properties, connect H: to \\FS01\Home\$\%username%.
6	Sign in on a client and confirm the drive maps automatically.

Validated Tests

Test	Account	Expected and final result
Automatic mapping	Sarah Lee (slee)	H: maps to \\FS01\Home\$\slee - Pass
Persistence	Sarah Lee	Test file remains after sign-out/sign-in - Pass
UNC mapping	Sarah Lee	net use H: reports status OK - Pass
Cross-user denial	Maria Lopez (mlopez)	Access to Sarah's folder is denied - Pass

Security boundary

The dollar sign hides Home\$ from casual browsing, but NTFS permissions provide the actual security boundary.

08

Folder Redirection

Documents stored centrally and available across clients

Setting	Final value
Server root	D:\Shares\RedirectedFolders
Hidden share	\\FS01\Redirected\$
GPO	VC-Folder-Redirection
Folder/method	Documents - Basic - create a folder for each user
Resulting path	\\FS01\Redirected\$\%USERNAME%\Documents
Security filter	Folder_Redirection_Users
GPO links	Finance, HR, IT, Operations, and Warehouse user OUs

Procedure

1	Create the Redirected\$ share and configure the root-folder permissions.
2	Create VC-Folder-Redirection and retain Authenticated Users with Read delegation.
3	Configure User Configuration / Policies / Windows Settings / Folder Redirection / Documents.
4	Select Basic, create a folder for each user, and enter \\FS01\Redirected\$ as the root path.
5	Enable exclusive rights and move existing Documents contents.
6	Leave contents in the redirected location if the policy is removed.
7	Link the GPO to all five department user OUs and filter it to Folder_Redirection_Users.

Validated Tests

- Anna Brown's Documents location resolves to \\FS01\Redirected\$\abrown\Documents.
- gpresult confirms VC-Folder-Redirection applied.
- A test document created on WIN11-01 appears on WIN11-02 with the same name, timestamp, and size.

09

PowerShell Logon Script

A safe GPO-deployed user-session marker

Setting	Final value
Script	VanFox-Logon.ps1
Central location	\\vanfox.local\SYSVOL\vanfox.local\scripts
GPO	VC-User-Logon-Script
Trigger/context	User sign-in / signed-in user
Output	%LOCALAPPDATA%\VanFox\LogonInfo.txt
Security filter	Authenticated Users

Final Script

```
$folder = Join-Path $env:LOCALAPPDATA "VanFox"
New-Item -Path $folder -ItemType Directory -Force | Out-Null

$details = @(
    "Last logon: $(Get-Date -Format 'yyyy-MM-dd HH:mm:ss')"
    "User: $env:USERDOMAIN\$env:USERNAME"
    "Computer: $env:COMPUTERNAME"
    "Home drive: $env:HOMEDRIVE"
)

$details | Set-Content -Path (Join-Path $folder "LogonInfo.txt")
```

Deployment and Validation

- 1 Save the script in the domain SYSVOL scripts directory.
- 2 Create VC-User-Logon-Script and configure it as a PowerShell logon script.
- 3 Link the GPO to all five department user OUs.
- 4 Run gpupdate /force, sign out, and sign back in.
- 5 Confirm LogonInfo.txt contains the current time, domain user, computer, and H: drive.
- 6 Run gpresult /r /scope:user and confirm the GPO is applied.

Validation account

Maria Lopez (mlopez) produced the expected LogonInfo.txt on WIN11-02, and gpresult confirmed delivery through VC-User-Logon-Script.

10

Group Policy Administration

Ten purpose-specific policies with clear targeting

GPO	Purpose	Scope / targeting
VC-Password-Policy	Password and account lockout	Domain level
VC-Company-Wallpaper	VanFox wallpaper	Department user OUs
VC-Map-Drives	S: department and P: Public	Item-level group targeting
VC-Folder-Redirection	Redirect Documents	Department OUs + filter group
VC-User-Logon-Script	Run VanFox-Logon.ps1	Department user OUs
VC-Deploy-Chrome	Assigned Chrome MSI	Managed computer OUs
VC-Deploy-Printers	Shared printer	Printer_Users
VC-Disable-USB-NonIT	Block removable storage	Non-exempt users
VC-LockScreen-5min	Lock after five minutes	Department user OUs
VC-LocalAdmins-ITOnly	Add authorized IT admins	Managed computer OUs

Security and User-Environment Settings

Policy	Final settings
Password	24 remembered; max age 60 days; min age 1 day; min length 14; complexity enabled; reversible encryption disabled
Account lockout	10 invalid attempts; 15-minute duration; reset after 15 minutes; administrator lockout enabled
Wallpaper	\\FS01\Public\Branding\VanFox-Wallpaper.png - style Fit
Lock screen	scrsave.scr; enabled; password protected; timeout 300 seconds

Resource Deployment Settings

Resource	Final configuration
Department drive	S: - Replace - reconnect - matching share group item-level targeting
Public drive	P: - \\FS01\Public - Replace - Public_Share_R targeting
Chrome	Assigned x64 MSI from \\FS01\Software\$\Chrome\googlechromestandaloneenterprise64.msi
Printer	Update \\FS01\VanFox-Printer - Printer_Users targeting - not default

Group Policy Security Controls

Control	Implemented behavior	Validation
USB restriction	All Removable Storage classes: Deny all access	Anna Brown receives Access Denied
USB exception	USB_Allowed has Read but is denied Apply Group Policy	Sarah Lee opens the USB device
Local administrators	GPP Update adds VANFOX\IT_Admins; existing required members are not deleted	net localgroup administrators lists IT_Admins
Folder Redirection	Folder_Redirection_Users security filtering	Anna Brown receives redirected Documents
Printer	Printer_Users item-level targeting	VanFox-Printer appears for authorized user

Administrative Standard

- Keep one administrative purpose per GPO.
- Link user settings to user OUs and computer settings to computer OUs.
- Use security filtering and item-level targeting for group-specific behavior.
- Run Group Policy Results or gpresult before changing link order, inheritance, or enforcement.
- Record purpose, scope, setting, test result, and rollback approach for each GPO.

Important distinction

VC-LocalAdmins-ITOnly is additive. It adds the IT_Admins domain group but does not remove built-in Administrator, Domain Admins, or other required existing members.

11

Client Join and End-to-End Validation

Two managed Windows clients receiving centralized services

Client	Address	Computer OU	Purpose
WIN11-01	10.0.2.30	VanFox\Computers\HR-PCs	HR managed client
WIN11-02	10.0.2.40	VanFox\Computers\IT-PCs	IT managed client

Join Procedure

1	Rename the computer to WIN11-01 or WIN11-02.
2	Assign the final static IPv4 address and set DNS to 10.0.2.15.
3	Confirm that dc01.vanfox.local and fs01.vanfox.local resolve internally.
4	Join vanfox.local with an authorized domain account and restart.
5	Move the computer object to HR-PCs or IT-PCs.
6	Run gpupdate /force and restart or sign out when requested.

Client Validation Commands

```
hostname
whoami
systeminfo | findstr /B /C:"Host Name" /C:"Domain"
ipconfig | findstr /I "IPv4"
ipconfig /all | findstr /I /C:"DNS Servers"
ping -4 -n 1 dc01.vanfox.local
ping -4 -n 1 fs01.vanfox.local
gpupdate /force
gpresult /r /scope:user
net localgroup administrators
```

Managed User Environment

Resource	Expected result
H:	Personal Home Folder for the signed-in user
S:	Department share selected through group targeting
P:	Read-only Public share
Documents	Redirected to the user's FS01 folder
Wallpaper	VanFox corporate background
Chrome and printer	Installed/connected from centralized policy

Final Validation Matrix

Validation test	Final result	Evidence location
Both clients registered in assigned AD OUs	PASS	Section 11 - Screenshot 01
WIN11-01 and WIN11-02 joined to vanfox.local	PASS	Section 11 - Screenshots 02-03
DC01 and FS01 resolve through internal DNS	PASS	Section 11 - Screenshots 02-03
Department/Public drives map; unauthorized access denied	PASS	Sections 06 and 11
Private H: drive maps and persists	PASS	Section 07
Cross-user Home Folder access denied	PASS	Section 07 - Screenshot 05
Documents redirects to FS01	PASS	Section 08 - Screenshots 01-04
Same redirected document appears on both clients	PASS	Section 08 - Screenshots 05-06
LogonInfo.txt is created by the logon GPO	PASS	Section 09
Company wallpaper and mapped drives apply	PASS	Section 11 - Screenshot 05
Chrome installs through assigned MSI	PASS	Section 11 - Screenshot 06
Shared printer appears for authorized user	PASS	Section 11 - Screenshot 07
Non-exempt user is blocked from USB storage	PASS	Section 11 - Screenshot 08
USB_Allowed user can access USB storage	PASS	Section 11 - Screenshot 09
IT_Admins appears in local Administrators	PASS	Section 11 - Screenshot 10
Authentication required after five minutes	PASS	Section 11 - Screenshot 11

Completion status

All implemented features in this guide were validated. Optional enhancements on the final page are not presented as completed work.

Troubleshooting and Change Record

The completed lab followed the revised design. The following implementation details and corrective actions were added while the environment was built and validated.

Observed condition	Cause / finding	Resolution or lesson
FS01 DNS queries timed out	DC01 was powered off	Start DC01 before validating dependent domain services
FS01 role installation failed	The session lacked administrative rights	Repeat the task with an authorized administrator account
fs01.vanfox.local returned NXDOMAIN	FS01 forward host resolution was not available	Verify/create the FS01 A record at 10.0.2.20 and retest
nslookup displayed Server: Unknown	The DNS server address lacked a reverse PTR record	Forward resolution remained functional; a reverse zone is an optional diagnostic improvement
New group behavior did not appear immediately	Existing user token did not include the new membership	Sign out and sign back in, then run gpupdate /force
Policy result uncertain	Scope, OU placement, or security filtering required confirmation	Check DNS, OU placement, group membership, gpresult, then the target resource

Added Final Implementation Details

- Recorded the exact 10.0.2.15, .20, .30, and .40 address plan.
- Added a dedicated 40 GB data volume, Access-Based Enumeration, and disabled offline caching on department shares.
- Added the Software\$ package source, final wallpaper path, and shared-printer name.
- Documented exact password, account-lockout, screen-lock, USB-exception, and local-administrator behavior.
- Combined validation and portfolio evidence into the 11-section GitHub structure.

Standard Troubleshooting Order

1	Confirm DC01 is running and the client uses 10.0.2.15 for DNS.
2	Resolve DC01, FS01, and vanfox.local by name.
3	Verify user and computer OU placement.
4	Verify security-group membership and refresh the user's sign-in token.
5	Run gpupdate /force and gpresult /r.
6	Inspect the mapped resource, application, printer, or restriction directly.
7	Check SMB, NTFS, and effective permissions when access is unexpected.
8	Review Event Viewer and change one variable at a time.

Portfolio Handoff

The GitHub repository is the evidence source for this guide. Each numbered section contains configuration notes, validation screenshots, and a short explanation of why the evidence matters.

github.com/vanneithawng/VanFox-Active-Directory-Lab

Final Checklist

Status	Completed item
PASS	Final DC01, FS01, WIN11-01, and WIN11-02 network settings recorded
PASS	Users, groups, and OUs match the implemented directory design
PASS	Home Folder permissions prevent cross-user access
PASS	Documents follows the same user between both clients
PASS	Logon script creates the expected local marker
PASS	Client evidence confirms intended Group Policy behavior
PASS	Validation results and troubleshooting notes are documented
PASS	Repository evidence contains no passwords or access tokens

Portfolio Summary

Project statement

Built and documented a four-VM Windows Server Active Directory lab with AD DS, DNS, organizational units, 20 user accounts, role-based security groups, domain-joined Windows 11 clients, secured department shares, private Home Folders, Documents Folder Redirection, PowerShell logon scripting, software and printer deployment, USB restrictions, and end-to-end Group Policy validation.

Optional Next-Level Enhancements

- Automate user, group, Home Folder, and permission creation with reviewed PowerShell.
- Add Windows LAPS for managed local administrator passwords.
- Add File Server Resource Manager quotas, file screening, and storage reports.
- Configure backup and perform a documented user-data restore test.
- Delegate limited password-reset and account-unlock permissions to IT_Helpdesk.

Scope statement

The optional enhancements above are future work. They are intentionally separated from the completed and validated implementation.